

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 24UMEC203L

B.E. III (ME)

6

PMT

SECOND B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(III SEMESTER)

3ME45A (ME) : PRODUCTION MACHINE

TOOLS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.



- (a) What are the various method available for taper-turning in a lathe? Explain their specific advantages and limitations. 5

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(Contd.)

- (b) Write a brief note on deep hole drilling operation. 4
- (c) What are the applications of shaper machines in a typical machine shop. Also compare shaper and planer in terms of the operation, and type of workpieces. 5
2. (a) Differentiate between up milling and down milling. Explain their applications mentioning the most commonly used method. 5
- (b) What are the various types of surface grinding approaches that are possible? Enumerate their advantages and applications. 5
- (c) Explain the functions served by a fixture. With the help of an example, explain why is foolproofing done fixtures. 4
3. (a) What are the purposes of fluxes and shielding gases in welding? 5
- (b) Explain thermit welding process. What products are welded by this method? 5
- (c) How do LBW and EBW differ from one another? 4

✓ 4. (a) Explain briefly various press working operations. 4

(b) Describe the following:

(i) Progressive dies 5

(ii) Compound dies

✓ 5. (c) What are the differences between automatic lathe and Capstan lathe? 5

(a) Briefly explain the principle of computer numerical control for machine tools. Mention the applications for which these are used. 5

(b) Describe the various types of control systems possible in CNC Operation. 5

(c) Explain the advantages and limitations of numerical control of machine tools. 4

✓ 6. Write short notes on any THREE of the following:

(i) Quick return mechanism

(ii) Balancing of grinding wheels

(iii) Plasma arc welding

(iv) Indexing - types and heads

(v) Tool layout

5+5+4=14